

A Technical Introduction to GPT language model

Section 1

Foundation models A foundation model is an AI model trained on broad data at scale such that it can be adapted to a wide range of downstream tasks. The most recent OpenAI's GPT-n series model is GPT-5. Other such models include Google's PaLM, a broad foundation model that has been compared to GPT-3 and has been made available to developers via an API, and Together's GPT-JT, which has been reported as the closest-performing open-source alternative to GPT-3 (and is derived from earlier open-source GPTs). Meta AI (formerly Facebook) also has a generative transformer-based foundational large language model, known as LLaMA. Foundational GPTs can also employ modalities other than text, for input and/or output. GPT-4 is a multi-modal LLM that is capable of processing text and image input (though its output is limited to text). Regarding multimodal output, some generative transformer-based models are used for text-to-image technologies such as diffusion and parallel decoding. Such kinds of models can serve as visual foundation models (VFMs) for developing downstream systems that can work with images.

Section 2

Evaluation and benchmarking The evaluation of generative pre-trained transformer models is carried out using a range of benchmarks and metrics, which aim to assess their performance across different tasks. Common approaches include accuracy on standard datasets, as well as other features such as robustness, bias, and toxicity. These models are typically tested on tasks such as natural language understanding, reasoning, answering queries, and code generation. Sometimes they combine multiple tasks to provide a broader assessment of model performance across domains. More recent approaches extend these evaluations to include other features such as fairness, efficiency, and transparency, in order to get a more accurate assessment of these models. Evaluation remains an active area of research, as existing tests may not accurately reflect real-world performance or the risks associated with large-scale generative models.

Section 3

A foundational GPT model can be further adapted to produce more targeted systems directed to specific tasks and/or subject-matter domains. Methods for such adaptation can include additional fine-tuning (beyond that done for the foundation model) as well as certain forms of prompt engineering. An important example of this is fine-tuning models to follow instructions, which is of course a fairly broad task but more targeted than a foundation

model. In January 2022, OpenAI introduced "InstructGPT" – a series of models which were fine-tuned to follow instructions using a combination of supervised training and reinforcement learning from human feedback (RLHF) on base GPT-3 language models. Advantages this had over the bare foundational models included higher accuracy, less negative/toxic sentiment, and generally better alignment with user needs. Hence, OpenAI began using this as the basis for its API service offerings. Other instruction-tuned models have been released by others, including a fully open version. Another (related) kind of task-specific models are chatbots, which engage in human-like conversation. In November 2022, OpenAI launched ChatGPT – an online chat interface powered by an instruction-tuned language model trained in a similar fashion to InstructGPT. They trained this model using RLHF, with human AI trainers providing conversations in which they played both the user and the AI, and mixed this new dialogue dataset with the InstructGPT dataset for a conversational format suitable for a chatbot. Other major chatbots currently include Microsoft's Bing Chat, which uses OpenAI's GPT-4 (as part of a broader close collaboration between OpenAI and Microsoft), and Google's competing chatbot Gemini (initially based on their LaMDA family of conversation-trained language models, with plans to switch to PaLM). Yet another kind of task that a GPT can be used for is the meta-task of generating its own instructions, like developing a series of prompts for 'itself' to be able to effectuate a more general goal given by a human user. This is known as an AI agent, and more specifically a recursive one because it uses results from its previous self-instructions to help it form its subsequent prompts; the first major example of this was Auto-GPT (which uses OpenAI's GPT models), and others have since been developed as well. Chain-of-thought reasoning is a prompting technique in which a language model generates intermediate reasoning steps before arriving at a final answer. This approach has been shown to improve performance on tasks requiring multi-step reasoning, such as mathematical problem solving and logical inference. By producing step-by-step explanations, the model is able to decompose complex problems into smaller, more manageable parts. This technique is often used in conjunction with few-shot prompting, where examples demonstrating the reasoning process are included in the input.

Section 4

Efficient transformer architectures The computational and memory requirements of transformer-based models increase significantly as they scale to larger sizes and longer input sequences. The standard self-attention mechanism has a quadratic complexity with respect to input sequence length, as described in Attention Is All You Need. Researchers proposed a number of efficiency improvements like sparse attention mechanisms and

memory-efficient architectures that reduce computational costs while supporting longer context windows. Models like BigBird, Reformer, and FlashAttention demonstrate structured attention patterns or optimized computation to improve scalability and efficiency. This has helped large language models to efficiently process long input sequences with reduced memory and computation during both training and inference.

Section 5

Emergent abilities Emergent abilities refer to capabilities that appear in large language models only when they reach a certain scale and are not present in smaller versions of the same models. These abilities are considered "emergent" because they arise as model size, training data, and compute increase. Examples of emergent abilities include multi-step reasoning, in-context learning (the ability to perform tasks based on examples provided in prompts without additional training), and improved performance on complex language and reasoning benchmarks. Research suggests that these capabilities do not scale linearly, but instead appear once models exceed certain thresholds in size and training scale. This phenomenon has influenced the development of larger GPT models and contributed to their increased effectiveness across a wide range of tasks.

Section 6

Ethical considerations and societal impact Generative pre-trained transformer models have raised a range of ethical and societal concerns, particularly regarding bias, misinformation, and environmental impact. Large language models can reproduce and amplify patterns present in their training data, including social biases, which may lead to unfair or misleading outputs. These models have also been associated with generating inaccurate or misleading information, as they are designed to produce fluent text rather than to verify factual accuracy. This has an impact on their use in applications such as automated content generation and information dissemination. Training large-scale models also requires huge computational resources, contributing to increased energy consumption and environmental costs. Concerns about the environmental impact of large AI systems have led to calls for more efficient training methods and more transparency in reporting resource usage.